Tutorial No : AMVS 1

Topic : Introduction to Air-conditioning Systems

1. Explain the main difference between a DX and a chilled water system. [4 points]

2. Sketch the heat and fluid flow schematic for an air-cooled chilled water system. [10 points]

3. Name the **equipment** and the **numbered components** shown in the Figure 3.1 to 3.4.

(a)

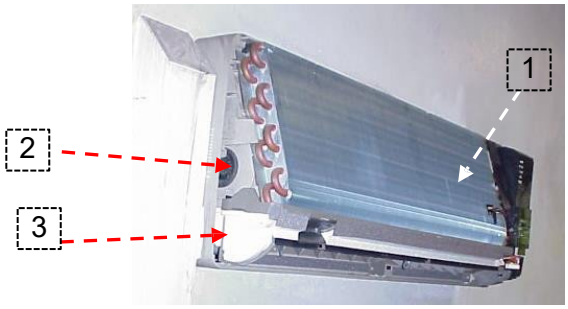


Figure 3.1 [2 points]

(b)

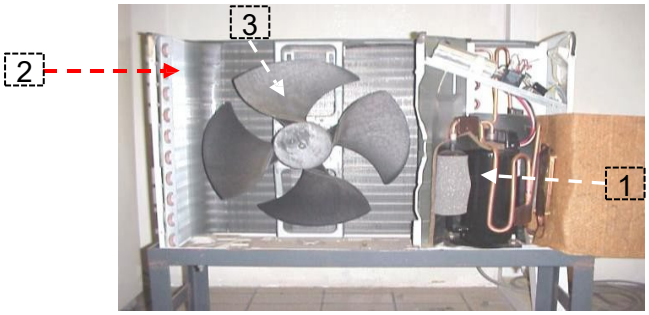


Figure 3.2 [2 points]

(c)

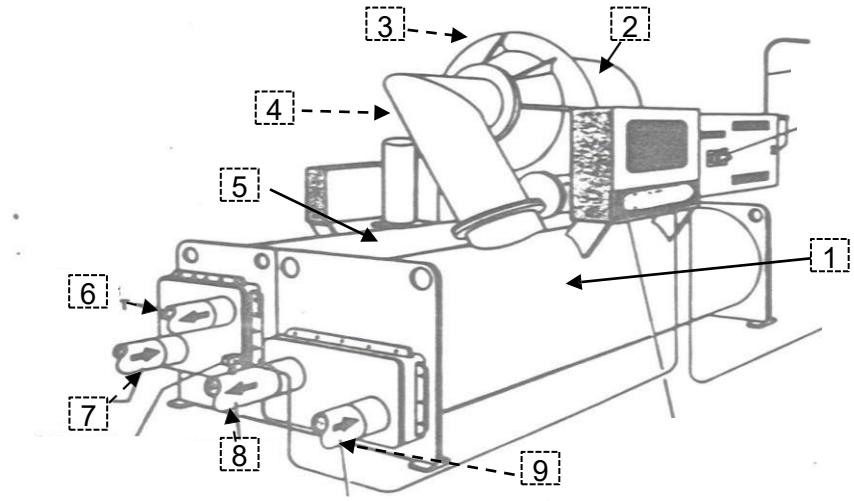


Figure 3.3 [5 points]

(d)

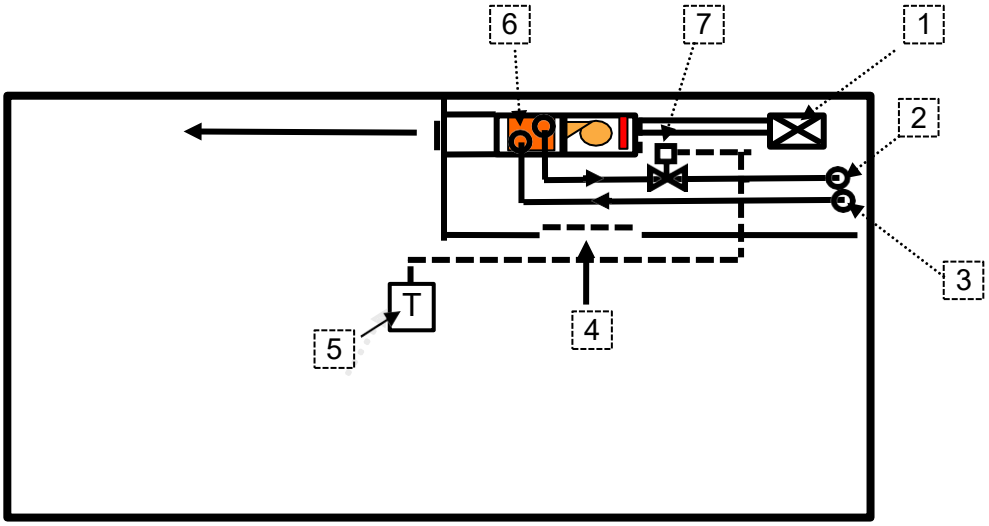


Figure 3.4 [4 points]

4. (a) Name the type of air-conditioning system shown in Figure 4 and equipment/ components numbered 1 to 7. [4 points]

(b) Describe how this system transfers heat from the air-conditioned space to the outdoor air. [7 points]

(c) The rated cooling capacity of the Component 1 is 18,000 Btu/h. What is its rating in RT? [1 point]

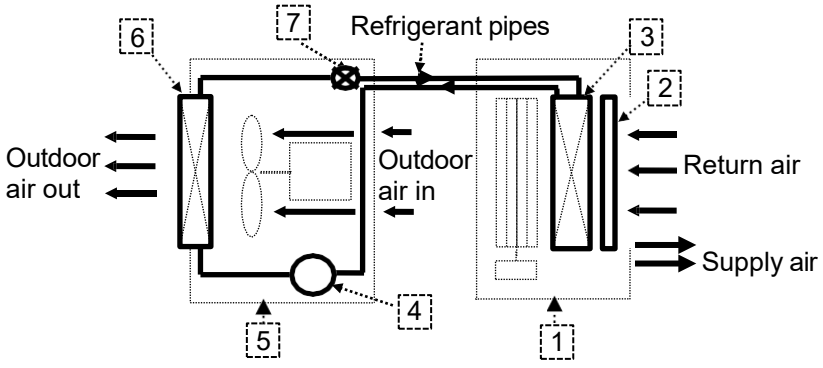


Figure 4

5. A water-cooled chiller consumes 480 kW_e when it provides a cooling capacity of 3600 kW_r. What is its efficiency in terms of kW_e/RT and COP? [3 points]

6. Describe the operation of a CAV system when the cooling demand decreases. Your description shall include how the various sensors operate to maintain the room air temperature at the setpoint value. [4 points]

7. Describe the operation of a conventional VAV AHU when the space cooling demand increases. Your description shall include how the various sensors operate to maintain the room air temperature at the setpoint value. [7 points]

8 (a) Name the type of air-conditioning system and equipment/components numbered 1 to 8 as shown in Figure 8. [5 points]

(b) Describe how heat is transferred from the air-conditioned spaces to the outdoors. Use the temperatures shown in the diagram in your description. [10 points]

(c) Explain the function of Component 8. [4 points]

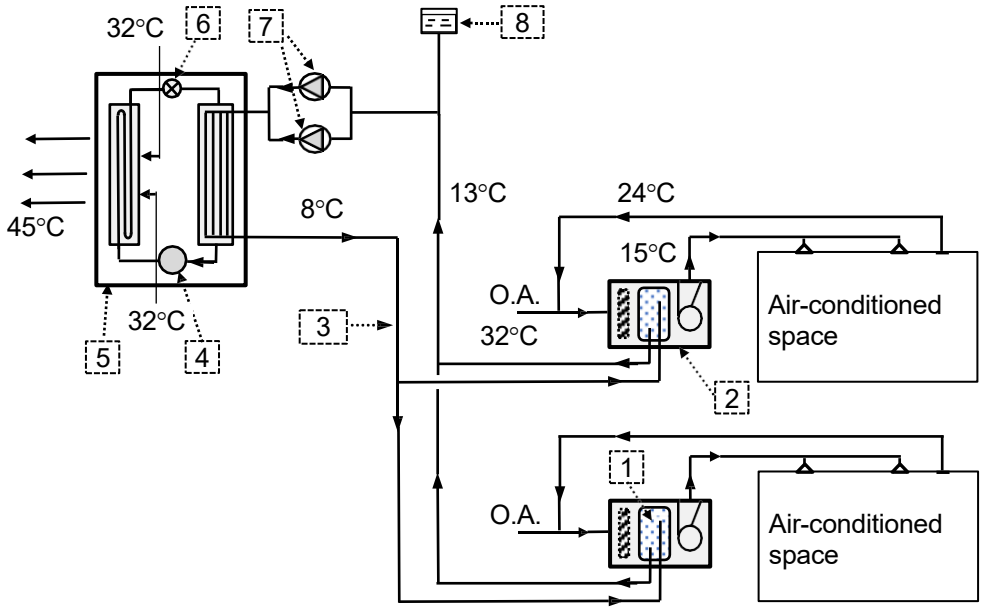


Figure 8